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IoT-BASED FUEL DISPENSER CARD

CASE STUDY: SOCIETE PETTOLIER -SP RWANDA

final project (dissertation) submitted to the faculty of Computing and Information Sciences in partial fulfillment of academic requirements for the award of a Bachelor's Degree of Science in **IT-Networking.**

Submitted by:

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Under the guidance of

Dr. Sumbiri Djuma

April, 2026

DECLARATION OF ORGINALITY

I do hereby declare that the work submitted in this dissertation is my own contribution to the best of my knowledge. The same work has never been submitted to any other University or institution. I therefore declare that this work is my own for the partial fulfillment of the award of a Bachelor's degree of science in IT-Networking at UNILAK.

Candidate's Name: ABAYISENGA J PHILIPPE

Candidate's signature: …………………………………………….

Submission date: ………………………………………………….

DEDICATION

This dissertation is dedicated to the people who have been my source of strength, inspiration, and love throughout this entire journey.

To my family: Thank you for being my unwavering support system. Your endless encouragement, sacrifices, and belief in me gave me the courage to keep pushing forward, even when things seemed impossible. Mom and Dad, your love has been my foundation, and I'm so grateful for every lesson you've taught me.

To my lecturers and mentors: Your wisdom and guidance have been invaluable. You challenged me, believed in my potential, and pushed me to reach heights I never thought possible. I couldn't have made it this far without your encouragement and trust in me.

And to my friends and peers: You've made this journey more meaningful, reminding me that even in the toughest moments, I'm not alone. Your support, laughter, and shared experiences have kept me going.

This dissertation is not just my work but a reflection of the love, care, and belief that has been poured into me by all of you. Thank you from the bottom of my heart.

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BONAFIDE CERTIFICATE

This is to certify that the project report entitled “**IoT-BASED FUEL DISPENSER CARD”** submitted by Abayisenga J Philippe (21681/2023) to the Faculty of Computing and Information Sciences at University of Lay Adventists of Kigali (UNILAK) in partial fulfillment to the requirements of the award of the bachelor’s degree of Science in IT-Networking is a record of bonafide work carried out by him under my guidance.

SUPERVISOR	DEAN OF FACULTY
Dr. Sumbiri Djuma	Dr. ENAN NYESHEJA MUHIRE PhD
Date: ……/…… /2026	Date: …… /…… /2026

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Finally, all glory and praise be to the Almighty God who provided strength, courage, and health that has enabled me to accomplish my studies at UNILAK

ABSTRACT

This project developed an IoT-Based Fuel Dispenser Card System for SP Rwanda Ltd to replace manual fuel sales recording. Using RFID cards and mobile money, customers can pay without cash, and sales are automatically recorded in real time. Managers and supervisors can view instant reports and monitor fuel stock. The system was tested at three branches and proved to be faster, more accurate, and more secure than the manual system. It reduces errors, prevents fraud, saves time, and improves customer experience.

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LIST OF ABBREVIATIONS

SP - Société Pétrolière Rwanda

DBMS - Database Management System

GUI - Graphical User Interface

HTML - Hyper Text Mark-up Language

CSS - Cascading Stylesheet

HTTP - Hyper Text Transfer Protocol

ICT - Information Communication Technology

IDE - Integrated Development Environment

IT - Information Technology

OS - Operating System

RAM - Random Access Memory

SQL - Structured Query Language

UML - Unified Modelling Language

OTP -One Time Password

AJAX - Asynchronous JavaScript and XML

CHAPTER 1: GENERAL INTRODUCTION

1.1 Introduction

Petrol is a vital natural resource, essential for the growing number of automobiles used worldwide. In Rwanda, security is seen as key to sustainable development, and the country has embraced Information Technology (IT) in various sectors, including security. IT plays an increasing role in daily life, allowing for online services, payments, and decision-making through management information systems (MIS), which enhance productivity.

However, petrol station management and reporting systems have lagged. By integrating IT, particularly MIS and RFID (Radio Frequency Identification) card transactions, SP petrol stations can enhance sales control, streamline operations, and improve security through contactless, cash-free payments. This technology offers secure, efficient solutions, helping operators reduce financial risks while better managing station activities.

1.2 Background of the study

Many services are emerging today, and technology is driving businesses to transition from paper-based systems to digital management. SP Petrol stations, a major global business, play a key role in supporting transportation. In Rwanda, several petrol station companies like SP Rwanda Ltd, ENGEN Ltd, and RUBIS Ltd supply various fuels (Premium, Gasoil, Kerosene) through multiple branches.

Currently, most SP petrol stations in Rwanda still use manual systems to record fuel sales, with attendants (pompistes) working in three shifts. They manually log fuel indices from pump machines and store tanks in "Report Books" after every shift. This process is time-consuming and prone to errors, as calculations must be done manually to track fuel sold and money earned. At the end of the month, sales data is consolidated in Excel and sent to headquarters.

The manual system also poses challenges for supervisors in maintaining accurate daily records. Stock ordering is limited to weekdays, which can lead to shortages on weekends. To address these inefficiencies, this project aims to transition petrol station operations from paper-based to digital systems, improving accuracy, speed, and management efficiency. The new system will enable supervisors to maintain daily records more easily and eliminate redundant manual work.

1.2 Problem Statement

SP Rwanda Ltd operates multiple branches across the country, each managed by a branch manager and a team leader who also serves as a fuel attendant. When the manager is absent, the team leader assumes managerial duties. Currently, petrol stations manually calculate daily fuel sales, with supervisors performing time-consuming calculations for each pump after every shift. Errors often occur due to human miscalculation, making it difficult to manage and retrieve data accurately and efficiently.

The current system also relies on Excel sheets, which can be corrupted, and lacks proper data security. Manual record-keeping slows decision-making and can lead to financial losses due to errors or fraud. To address these challenges, the proposed " IoT-Based Fuel dispenser card " will provide a digital solution. It will enable supervisors to monitor fuel sales automatically, generate accurate records, and improve overall data security, ensuring smoother and more efficient management for SP Rwanda Ltd.

1.3 Objectives

1.3.1 The main objective

the main objective of this study is to develop an IoT-Based Fuel dispenser card system utilizing RFID Cards for the transaction

1.3.2 Specific Objectives

i. To design an interface that will allow users, such as fuel attendants and supervisors, to interact with the system efficiently.

ii. To design a database that keeps accurate and comprehensive fuel sales records, including details of card transactions and fuel stock levels.

iii. To implement a system that generates detailed fuel sales reports for supervisors and managers, providing accurate data on sales and stock.

1.4 Research Questions

i. What are the main gaps in the current fuel management system at SP Rwanda Ltd?

ii. How can RFID-based IoT systems improve transaction accuracy and security?

iii. What technological features can simplify fuel sales recording and reduce delays?

iv. How can the system ensure accessibility for users with varying technical skills?

1.5 Expected Results

The IoT-Based Fuel Dispenser Card System is expected to transform fuel management by making it faster, more transparent, and more secure. Applicants (customers) will benefit from cashless payments via RFID cards, while attendants will experience reduced manual work. Supervisors and managers will gain access to real-time sales data and automated reports, improving operational oversight and reducing fraud.

1.6 Choice of the Study

The study focuses on SP Rwanda Ltd due to its strategic role in Rwanda's petroleum sector and the persistent challenges its branches face with manual systems. By offering a tailored IoT-based solution, this project contributes to Rwanda's digital transformation goals in the energy sector.

1.7 Significance of the Study

This study offers practical benefits, including:

Improving Transparency and Trust: The system provides accurate, real-time records for both station managers and customers Supporting Rwanda's Digital Economy Goals: Aligns with national efforts to adopt IoT and cashless payment technologies. Serving as a Model for Other Petrol Stations: The system can be replicated by other fuel companies in Rwanda and beyond.

1.8 Scope and Limitations of the Study

IoT-Based Fuel dispenser card system is designed for use in petrol stations, focusing on automating the sales and reporting processes. The study will be conducted at selected SP Rwanda Ltd branches from October 2025 to January 2026. The system will facilitate card-based transactions, allowing customers to pay with RFID cards. However, the system may face limitations in areas with weak network infrastructure, as internet connectivity is required to sync sales data and generate reports in real-time. Additionally, SP petrol stations without card reader machines may face hardware compatibility issues.

1.8 Research Methods & Techniques

1.8.1 Information Collection Techniques

Interviews: Conducted with branch managers, supervisors, and fuel attendants to understand current challenges.

Surveys and Questionnaires: Distributed to staff and customers to gather feedback on desired features.

Document Analysis: Reviewed existing fuel sales logs and Excel-based reports to identify inefficiencies.

1.8.2 Design Methods

Prototyping: Created wireframes of the system interface and improved them iteratively based on user feedback.

Modular Development: The system was divided into independent modules (e.g., user login, RFID payment, reporting) for easier testing and integration.

Agile Development: The system was built in iterative sprints, allowing continuous improvement and adaptation.

User-Centered Design: The interface was designed to be simple and accessible for users with limited technical experience.

1.9 Organization of the Study

This dissertation is divided into six chapters:

Chapter 1: General Introduction – Presents the background, problem statement, objectives, scope, and methodology.

Chapter 2: Literature Review – Reviews related works, technologies, and compares existing systems.

Chapter 3: System Analysis and Methodology – Describes data collection, development approach, and tools used.

Chapter 4: System Design – Includes UML diagrams, database design, and system architecture.

Chapter 5: System Implementation and Testing – Discusses coding, screenshots, testing strategies, and evaluation.

Chapter 6: Conclusion and Recommendations – Summarizes findings, limitations, and future improvements.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This chapter reviews various research studies and literature from both local and international sources relevant to fuel management systems and the use of RFID (Radio Frequency Identification) cards for transactions. The literature includes books, journals, articles, electronic materials such as e-books, and dissertations that align with the study's objectives.

2.2 Definition of Key Terms

2.2.1RFIDTechnology

RFID (Radio Frequency Identification) technology uses electromagnetic fields to automatically identify and track tags attached to objects. In this context, RFID cards are used for cashless transactions at petrol stations, offering a convenient and secure payment method. The use of RFID cards in the Smart Petrol Station Management System enhances accuracy, security, and efficiency by reducing the need for cash transactions and minimizing human error.

According to Gupta and Malik (2005), a Management Information System (MIS) is a computer-based system providing flexible and fast access to accurate data. MIS supports management by generating useful information for decision-making. In the context of petrol stations, MIS helps in managing sales, inventory, and reporting processes, allowing for better decision-making and operational control.

2.2.3 System

Alexander (2015) defines a system as a set of interacting or interdependent components forming a complex whole. In the Smart Petrol Station Management System, various components, such as fuel pumps, RFID card readers, and the database, work together to automate the fuel sales process, improving efficiency and accuracy.

2.2.4 Information System

An information system integrates components for collecting, storing, and processing data to provide valuable information. Business firms use information systems to manage operations and interact with customers. In the Smart Petrol Station Management System, the information system handles fuel sales, manages transactions via RFID cards, and provides data for reporting and analysis, helping managers make informed decisions.

2.2.5 Web Application

Web applications are software systems accessible through a web browser over a network. The Smart Petrol Station Management System is a web application that allows managers, supervisors, and pompistes to access real-time sales data, transaction history, and inventory management through a secure online platform. This allows for easier monitoring of multiple branches and improved data accessibility.

2.2.6 Database Management System(DBMS)

A Database Management System (DBMS) is software that defines, manipulates, retrieves, and manages data in a database. In the Smart Petrol Station Management System, the DBMS is responsible for securely storing transaction data, fuel stock levels, and customer information, while also supporting queries and generating reports on fuel sales and usage.

2.3 Proposed System Vs Current System

This section compares the current manual fuel sales system with the proposed Smart Petrol Station Management System, which integrates RFID cards for transactions. The comparison highlights the key differences between the two systems.

2.3.1 Current System:

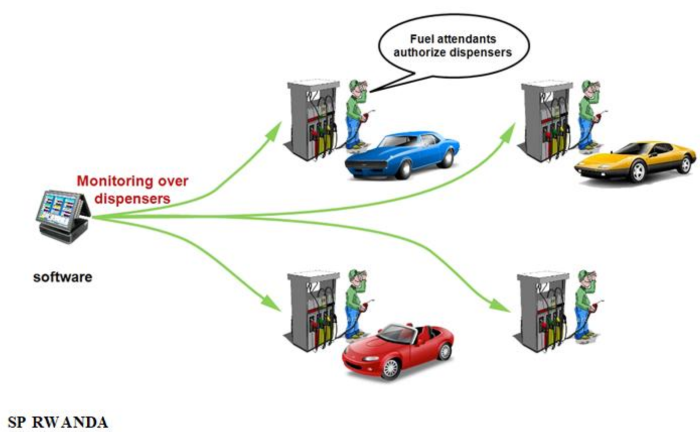


Figure 1 System Analysis

The current manual system in petrol stations involves recording fuel sales and transactions by hand using paper or spreadsheets. This method is time-consuming, prone to human errors, lacks real-time data, and requires manual verification of sales and stock. Managing records across multiple branches is inefficient, and the risk of data loss or corruption is high.

2.3.2 Proposed System

The proposed Smart Petrol Station Management System automates fuel sales through RFID card transactions. This system provides real-time sales data, reduces errors, and ensures secure transactions. With automatic record-keeping, the system streamlines operations, allowing supervisors to monitor sales and inventory easily. The RFID cards also offer a cashless, secure method of payment, minimizing fraud and manual errors. While the system requires a higher initial setup cost and technical expertise, the long-term benefits include improved efficiency, data security, and better decision-making.

2.3.3 Locally and International Comparison

The proposed system offers significant advantages over the current manual system, including better accuracy, time efficiency, and real-time access to sales and inventory data. It also provides enhanced security, reduced fraud, and scalability for multiple branches. However, it involves higher initial setup costs and technical infrastructure, requiring training for staff and technical support for seamless integration.

Countries like Kenya and South Africa have adopted RFID-based fuel management systems, reducing fraud and improving efficiency. Rwanda can benefit from similar technology, tailored to local infrastructure and user needs.

2.3.4 Personal Contribution

Several studies have explored the benefits of using RFID technology for transaction management in various sectors, showing its ability to streamline processes, reduce errors, and enhance security. In the context of petrol stations, RFID-based systems provide a more efficient way of managing fuel sales compared to manual methods. Studies have demonstrated that RFID technology improves accuracy in tracking sales, prevents fraud, and simplifies the reporting process for managers. The Smart Petrol Station Management System adopts these benefits to address the inefficiencies in manual fuel sales tracking, offering a modern solution that ensures secure, cashless transactions and real-time data access.

CHAPTER THREE: RESEARCH METHODOLOGY

3.1 Introduction

This chapter describes the methodology used to develop the Smart Petrol Station Management System. The Waterfall model is applied for this project, which ensures a sequential approach to system development. The chapter also details the data collection methods, software technologies, and how RFID cards are integrated into the system for secure transactions.

3.1 Data Collection and Procedures Used

Data was collected through various methods including documents, interviews, questionnaires, and field observations. These methods were designed to capture both user requirements and technical needs.

3.2 Primary Data

3.2.1 Documentation

We reviewed literature and documentation related to petrol station management, RFID card technology, and data security. Existing systems and their shortcomings were analyzed to understand how to improve operations with RFID technology for faster, more secure transactions.

3.2.2 Interviews

Interviews were conducted with managers, pompistes, and team leaders at different petrol stations. The interviews provided insights into the manual processes currently in place and how RFID cards could improve efficiency and security.

3.2.3 Questionnaires

Questionnaires were distributed to gather data from petrol station workers and managers regarding their experiences with the current system and their expectations for a digital management system.

3.2.4 Observations

We conducted site visits to petrol stations to observe the manual transaction processes and reporting methods. These observations helped in understanding how RFID card technology could streamline processes and reduce errors.

3.3 Secondary Data

Secondary data was collected from academic journals, online resources, and technical documents that provided insights into RFID card usage, system security, and petrol station operations. This data informed the design of the system and ensured it aligned with best practices in technology and management.

3.4 Software Development Life Cycle

The Waterfall model was selected for the development of the Smart Petrol Station Management System. This methodology allows for a structured approach where each phase must be completed before the next begins. The stages include Requirements Gathering, System Design, Implementation, Testing, and Maintenance.

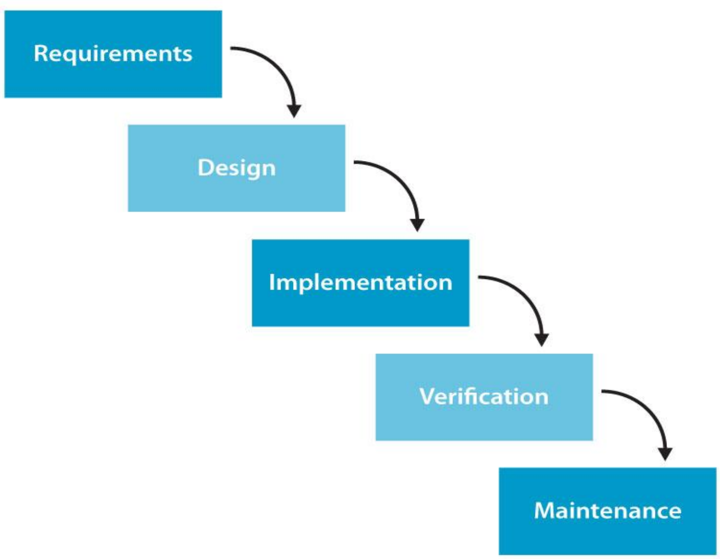


Figure 2:Waterfall model

3.4.1 Requirements Gathering

In this phase, we identified the functional and non-functional requirements of the system. Discussions with stakeholders highlighted the need for RFID card integration to improve transaction accuracy and security. The system also needed to support reporting functionalities for managers.

3.4.2 System Design

The system was designed to include modules for recording fuel sales, generating reports, and processing payments via RFID cards. The database structure was carefully designed to store transaction details securely. The design also incorporated OTP (One-Time Password) generation for additional payment security

3.4.3 Implementation

The coding phase involved developing the system using PHP for the backend, with RFID card readers integrated into the transaction modules. JavaScript and CSS were used for the frontend, ensuring an intuitive user interface. RFID cards were used to authenticate and authorize payments, and the system securely recorded all transactions in a MySQL database.

3.4.4 Testing

The system underwent rigorous testing, including unit testing for individual features like RFID card transactions, integration testing to ensure seamless interaction between modules, and user acceptance testing to ensure the system met the users' needs. Security testing was also conducted to verify that RFID card data and transaction records were safe from breaches.

3.4.5 Maintenance

After deployment, the system will be regularly monitored to ensure its smooth operation. The use of RFID cards in transactions will be continuously optimized to prevent any potential issues related to card recognition or payment errors. Routine updates and bug fixes will be made as necessary

3.5 Software Technologies

3.5.1 Backend

- MySQL: Used to store all transaction data securely, including RFID card details, fuel sales, and payment records.

3.5.2 Frontend

- PHP: Serves as the main scripting language to manage transactions, payments, and system interactions with RFID cards.
- JavaScript: Provides interactive elements for the user interface, including fuel price displays and transaction processing.
- CSS: Ensures the system has an easy-to-use and responsive design for both pompistes and managers.

CHAPTER FOUR: ANALYSIS AND DESIGN

4.1 Introduction

The requirements analysis phase is critical for the successful development of the Smart Petrol Station Management System. It involves identifying and understanding user needs, technical specifications, and the constraints within which the system will operate. This chapter outlines the functional and non-functional requirements of the system, along with various models to visually represent these requirements.

4.2 Context Diagram

The context diagram provides a high-level overview of the Smart Petrol Station Management System, illustrating how the system interacts with external entities. It identifies the key actors involved, such as petrol station managers, pompistes, customers, and the payment processing system (including RFID card readers). The diagram shows the flow of information between these actors and the system, helping to define the boundaries of the project.

- **Actors**:
    - **Petrol Station Manager**: Oversees operations, monitors sales reports, and manages pompistes.
    - **Pompiste**: Handles fuel dispensing, records sales, and uses RFID cards for transactions.
    - **Customer**: Uses RFID cards to make payments for fuel.
    - **Payment Processing System**: Authenticates transactions and processes payments via RFID cards.

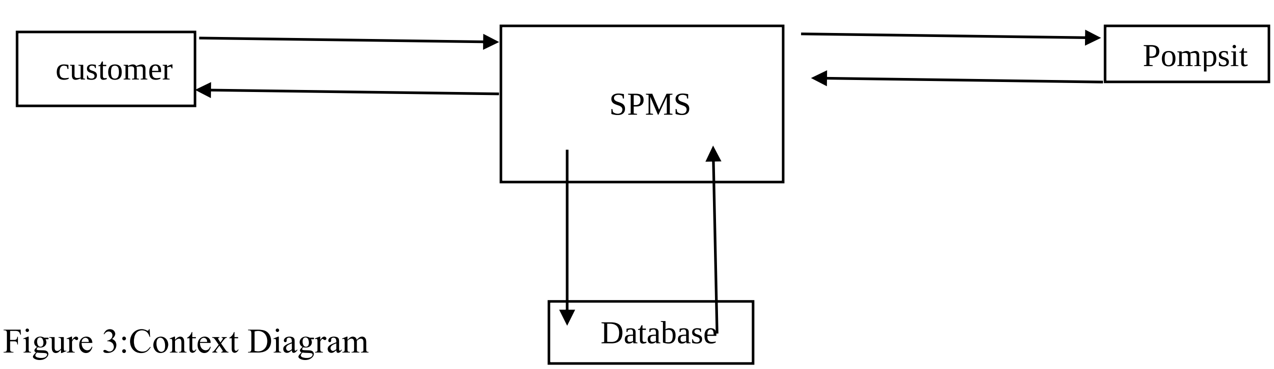


Figure 3:Context Diagram

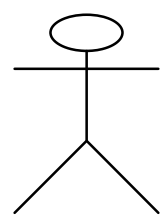
4.2 Use Case Diagram

4.2.1 Introduction

Use Case diagram shows a set of actors and Use Case, and their relationships, addresses static Use Case of the system; and also, is important for organizing and modeling the external behavior of the system.

Use case tools description:

Table 1: Use Case Tools Description

An actor: ❖ Is a person or system that derives benefit from and is external to the subject? ❖ Is depicted as either a stick figure (default) or, if a nonhuman actor is involved, a rectangle with “Actor” in it (alternative). ❖ Is labeled with its role. ❖ Can be associated with other actors using a specialization/super class association, denoted by an arrow with a hollow arrowhead. Is placed outside the subject boundary.	 Actor Name

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Use Case

A use case is a list of actions or event steps, typically defining the interactions between a role (known in the Unified Modelling Language as an actor) and a system, to achieve a goal and is represented as follows:
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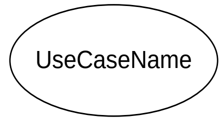

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A subject boundary:

Includes the name of the subject inside or on top.
Represents the scope of the subject,
e.g., a system or an individual business process.

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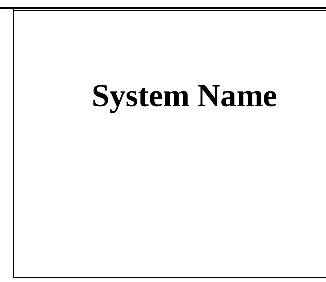

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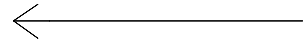
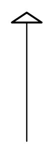
An include relationship: ❖ Represents the inclusion of the functionality of one-use case within another. ❖ Has an arrow drawn from the base use case to the used Use Case?	
An extend relationship: ❖ Represents the extension of the Use Case to include optional behavior. ❖ Has an arrow drawn from the extension use case to the base Use Case?	
A generalization relationship: ❖ Represents a specialized use case to a more generalized one. ❖ Has an arrow drawn from the specialized use case to the base use case?	
An association relationship: ❖ Links an actor with the Use Case (s) with which it interacts.	

4.2.2 Use Case Diagram Description

The use case diagram represents the actors and what they will work on as its meaning process.

Figure 4:Use Case Diagram

Use case description

Use Case description details what a use case do, and what it requests in order to be well executed.

Each use case looks like this:

- **Name:** a name of a use case
- **Description:** what a system intends to do
- **Actor:** the actor involved in the use case

- **Pre-condition:** the system state before the use case can begin
- **Post-condition:** the system state when the use case is over
- **Normal flow**: the actual steps of the use case
- **Alternative flow:** steps which may happen in case a normal flow fails.

Descriptions of the use cases:

Table 2: Manager Login

Name: Login
Actor: Manager
Description: This use case allows manager to login to the system for viewing the supervisors and their reports.
Pre-condition: he should have access to the system means has credentials.
Post-condition: If the use case is successfully allowed manager to login, he/she will access the system.
Normal Flow: 1. Manager will login into the system with his/her username and password 2. Confirmation message of login will be displayed. 3. The system checks the information provided if true the account is created. Then the actor is able to fill the username and password. 4. System validates names and password, and if finds correct allow the actor to log into the system.
Alternative flow: • if the information is not valid the system shows the error message. • The actor can decide to go back to the beginning of the main flow or cancel the login, at that point the use case ends.

Table 3 : Register Supervisor

Name: Register Supervisors
Actor: Manager
Description: This use case allows manager to register supervisors in order to give the access to control daily records activity and managing sales done.
Pre-condition: he should have access to register new supervisors and assigning them to his branch.
Post-condition: If the use case is successfully allowed manager to register supervisor, he/she will receive a notification message of successful or failure.
Normal Flow: 1. Manager will login into the system with his/her username and password 2. Request registration form. 3. System provides the registration form. 4. Manager send the information. 5. The system checks the information provided if true the account is created. Then the actor is able to fill the username and password. 6. System validates names and password, and if finds correct allow the actor to log into the system. 7. Successful message displayed.
Alternative flow: • if the information is not valid the system shows the error message. • Manager can decide to go back to the beginning of the main flow fill the form and resends.

Table 4: Register Pompiste

Name: Register Pompiste
Actor: Supervisor
Description: This use case allows supervisor to register pompiste to the system in order to make some records of fuel sold
Pre-condition: supervisor should have access to register pompiste.
Post-condition: If the use case is successfully allowed supervisor to register pompiste, he/she will receive a notification message of successful or failure.
Normal Flow: 1. Supervisor will login into the system with his/her username and password 2. Request registration form. 3. System provides the registration form. 4. Supervisor send the information Successful message displayed.
Alternative flow: • If the information is not valid the system shows the error message. • Manager can decide to go back to the beginning of the main flow fill the form and resends.

Table 5: Recording Fuel Sold

Name: Record Sales
Actor: Pompiste
Description: This use case allows pompiste to records fuel sold
Pre-condition: pompiste must have access to records sales
Post-condition: If the use case is successfully allowed supervisor to register pompiste, he/she will receive a notification message of successful or failure.
Normal Flow: 1. Pompiste will login into the system with his/her username and password 2. Add client's consummation. 3. System will provide him fuel prices and quantity. If fuels still in the tank task will proceed 4. System will send the OTP code to the client to make payment. 5. Client will make payment 6. Pompiste will approve payment. 7. Information will be sent automatically and updates make changes to UI and database with the remaining fuels and out liters with costs calculations. 8. Successful message displayed.
Alternative flow: • If the information is not valid the system shows the error message. • Pompiste can decide to go back to the beginning of the main flow fill the form and resends clients records of fuel sold.

Table 6: View Sales

Name: View Sales
Actor: Branch Manager, Supervisor and Pompiste
Description: This use case allows actors to view sales and show up information the one makes sales.
Pre-condition: all must have access to view sales
Post-condition: If the use case is successfully allowed actors to view sales.
Normal Flow: 9. Actors will login into the system with his/her username and password 10. System will provide sales details. 11. Successful message displayed.

4.3 Class diagram

A class Diagram is a type of static structure diagram that describes the structure of the system by showing the system's classes, their attributes, operations (or methods) and the relationships among objects. Class diagram shows a set of classes, interfaces and collaborations and their relationships; addresses static design view of a system.

The followings are components of a class diagram:

4.3.1 Classes

Classes represent an abstraction of entities with common characteristics. Associations represent the relationships between classes.

Illustrate classes with rectangles divided into compartments. Place the name of the class in the first partition (centred, bolded, and capitalized), list the attributes in the second partition (left-aligned, not bolded, and lowercase), and write operations into the third.

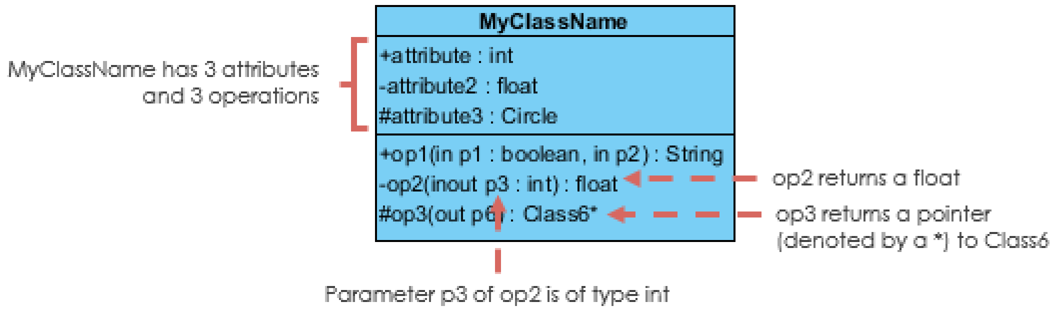


Figure 5:Classes

4.3.1.1 Active Classes

Active classes initiate and control the flow of activity, while passive classes store data and serve other classes. Illustrate active classes with a thicker border.

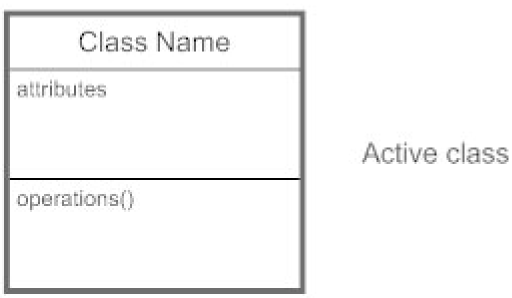
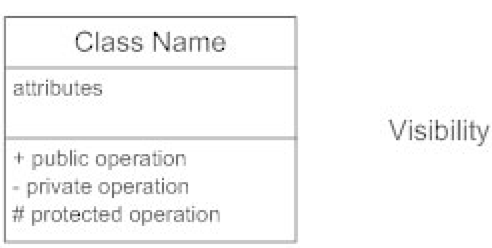


Figure 6: Classes

4.3.1.2 Visibility

Use visibility markers to signify who can access the information contained within a class. Private visibility, denoted with a - sign, hides information from anything outside the class partition. Public visibility, denoted with a + sign, allows all other classes to view the marked information. Protected visibility, denoted with a # sign, allows child classes to access information they inherited from a parent class.



Marker	Visibility
+	public
-	private
#	protected
~	package

Figure 7:Visibility

4.3.2 Class diagram Details

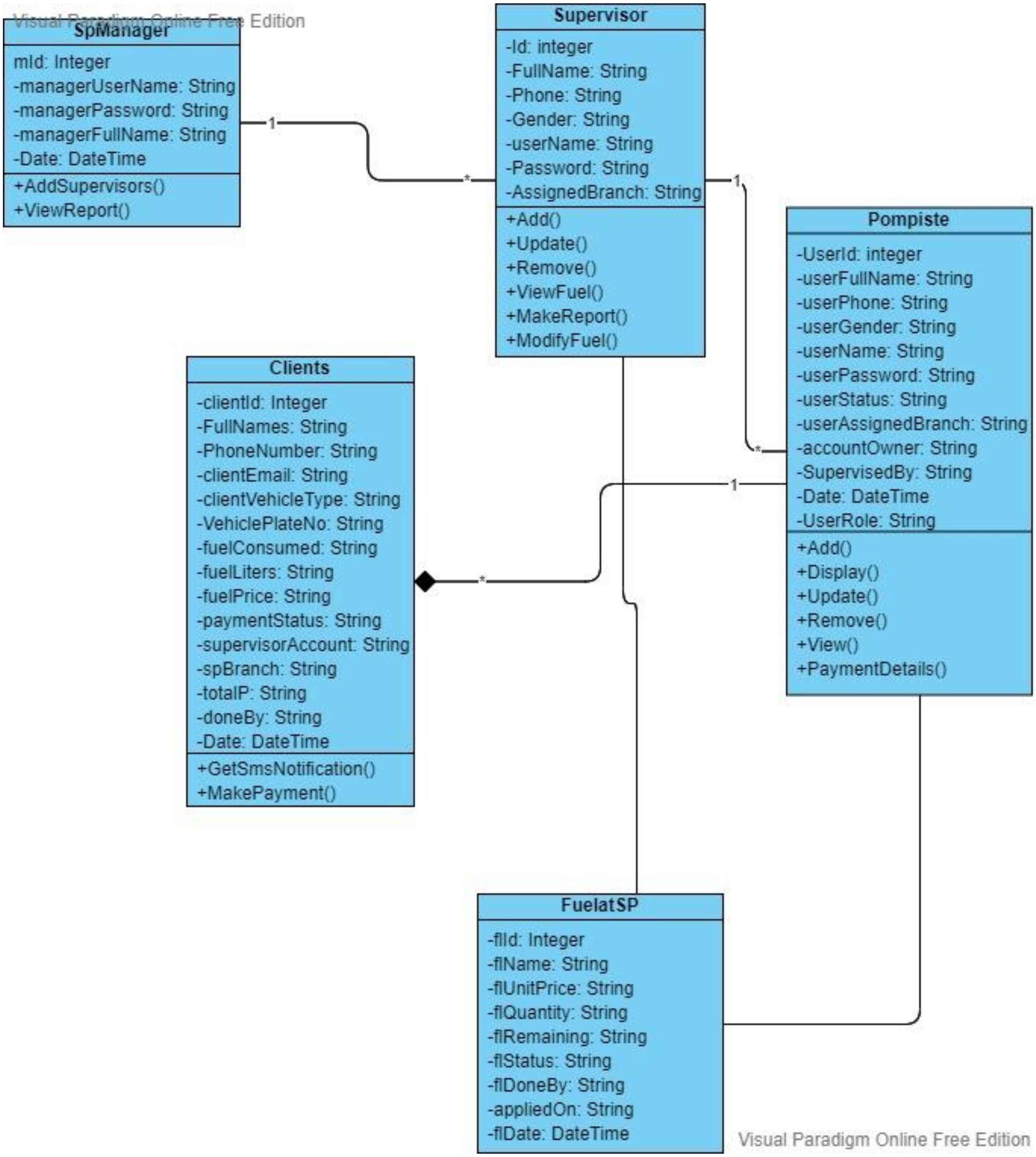


Figure 8: Class Diagram

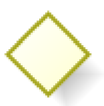
4.4 Activity Diagram

Activity diagrams are mostly graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency. In the Unified Modelling Language, activity diagrams are intended to model both computational and organizational processes (i.e. workflows). Activity diagrams show the overall flow of control.

Activity diagrams are constructed from a limited number of shapes, connected with arrows.

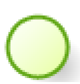
The most important shape types:

A task is an atomic activity that is included within the process

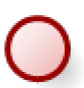


Exclusive gate ways (decisions) are locations within a business process where the sequence can take two or more alternative paths

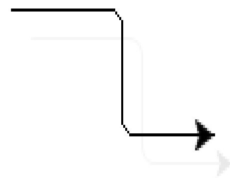
Bars represent the start (split) or end (join) of concurrent activities



The start event indicates where a particular process will start.



The End event indicates where a particular process will End.



A Sequence diagram or flow is used to show the order that activities will be performed in a process.

Activity diagrams may be provided as a form of flowchart. Typical flowchart techniques lack constructs for expressing concurrency. However, the join and split symbols in activity diagrams only resolve this for simple cases; the meaning of the model is not clear when they are arbitrarily combined with decisions or loops.

Admin login activity diagram

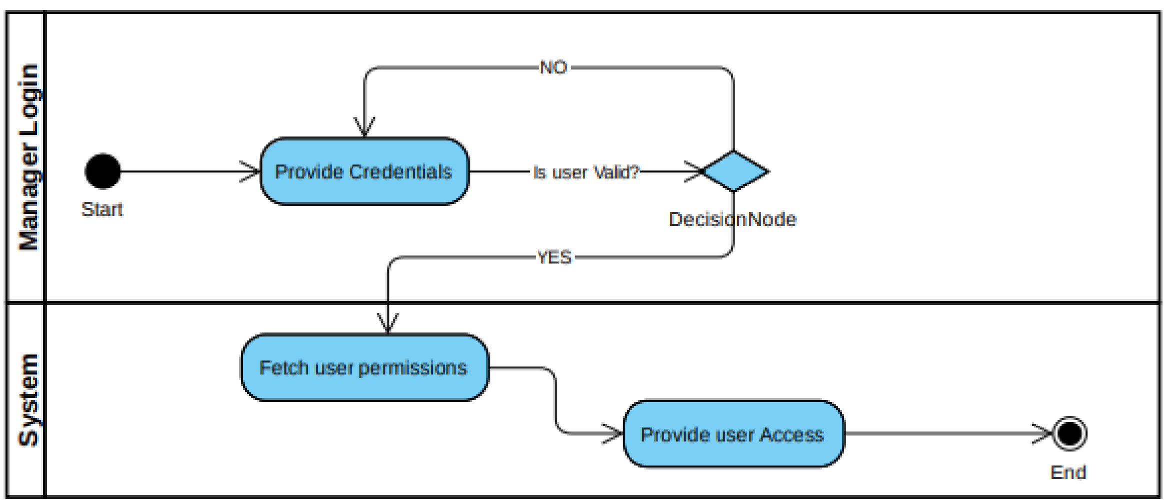


Figure 9:Activity Diagram of manager login

Supervisor Credentials Activity Diagram

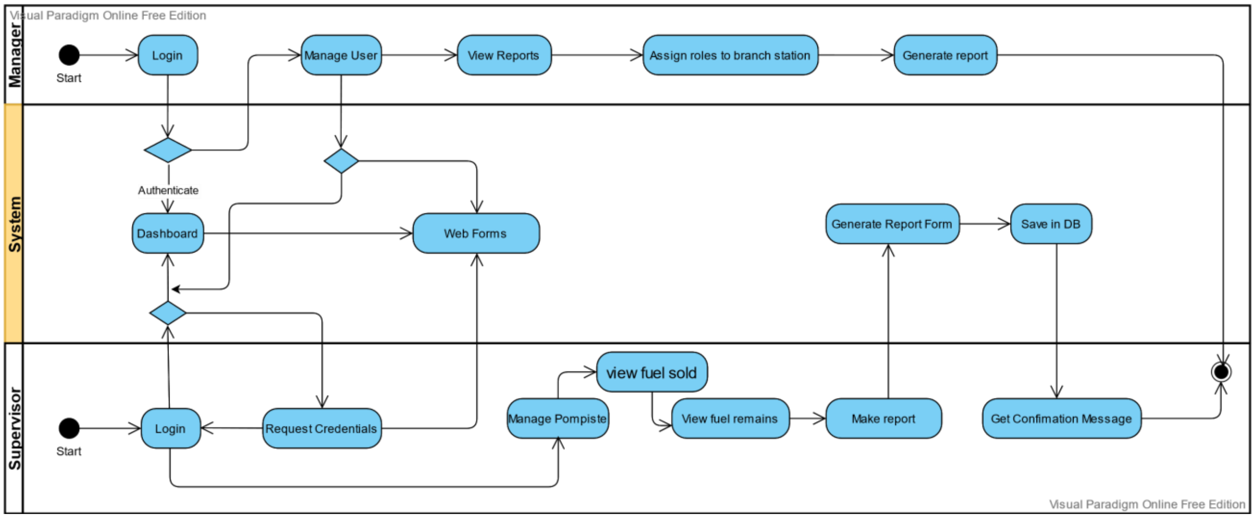


Figure 10:Activity Diagram of supervisor details

Supervisor Management Activity Diagram's Details

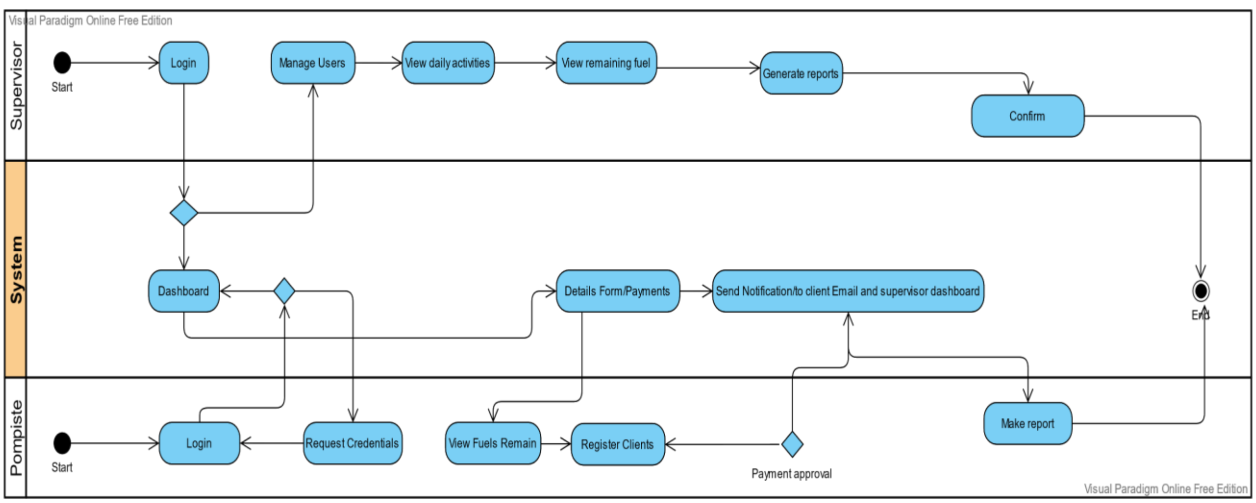


Figure 11:Activity Diagram of supervisor control details

CHAPTER FIVE: SYSTEM IMPLEMENTATION AND TESTING

5.1 Introduction

System design translates the requirements into a blueprint for building the system. This chapter covers the architectural design, user interface design, database design, and security measures.

Presentation of the New System

This section illustrates the results of the implementation of this system. These are the user interfaces or web pages that result from running the codes. Online Request Newspapers interfaces and display pages. Some of them are displayed in form of the screenshots but few are illustrated in the following section:

5.2 Home page

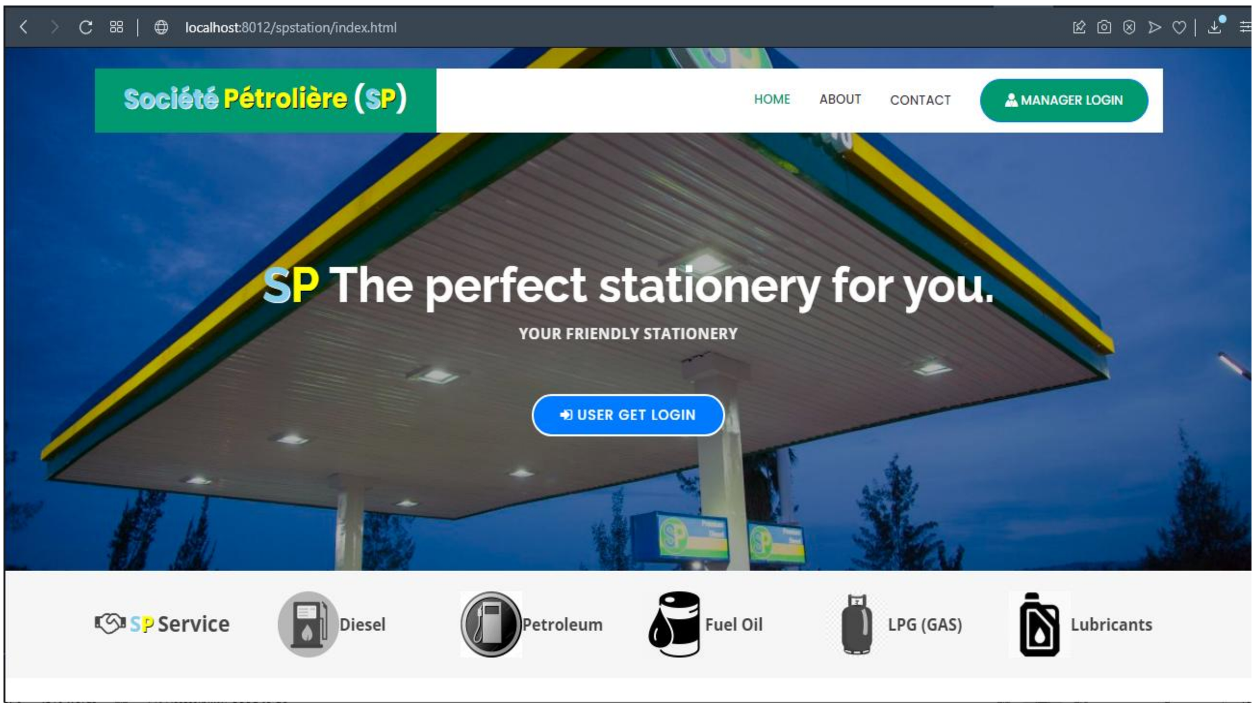


Figure 12: System home page

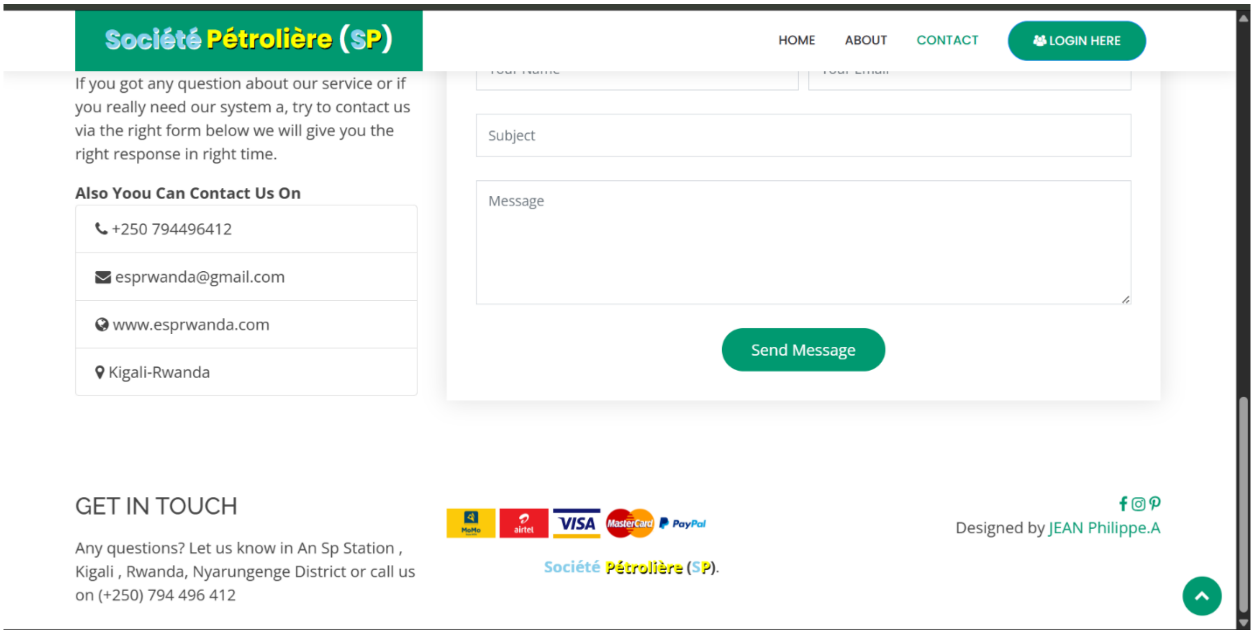


Figure 13: System home page

5.3 Manager login form

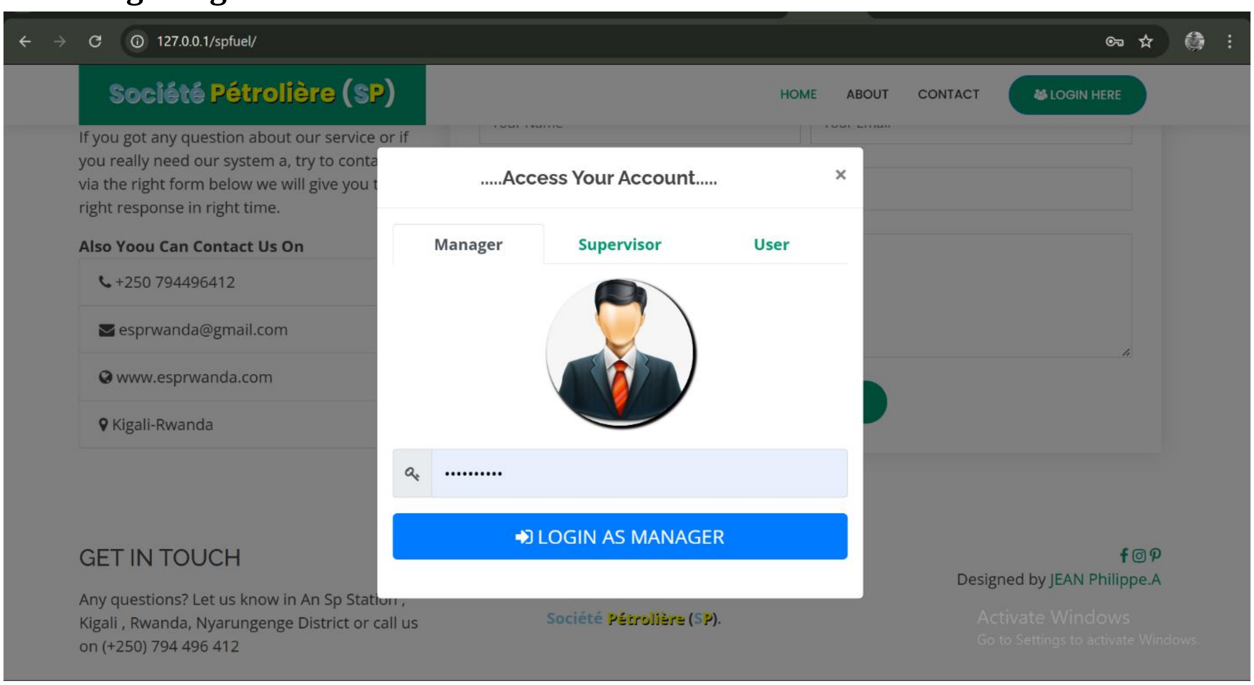


Figure 14: Manager login form

5.3.1 Manager will be able to manager supervisor by giving them access to the system

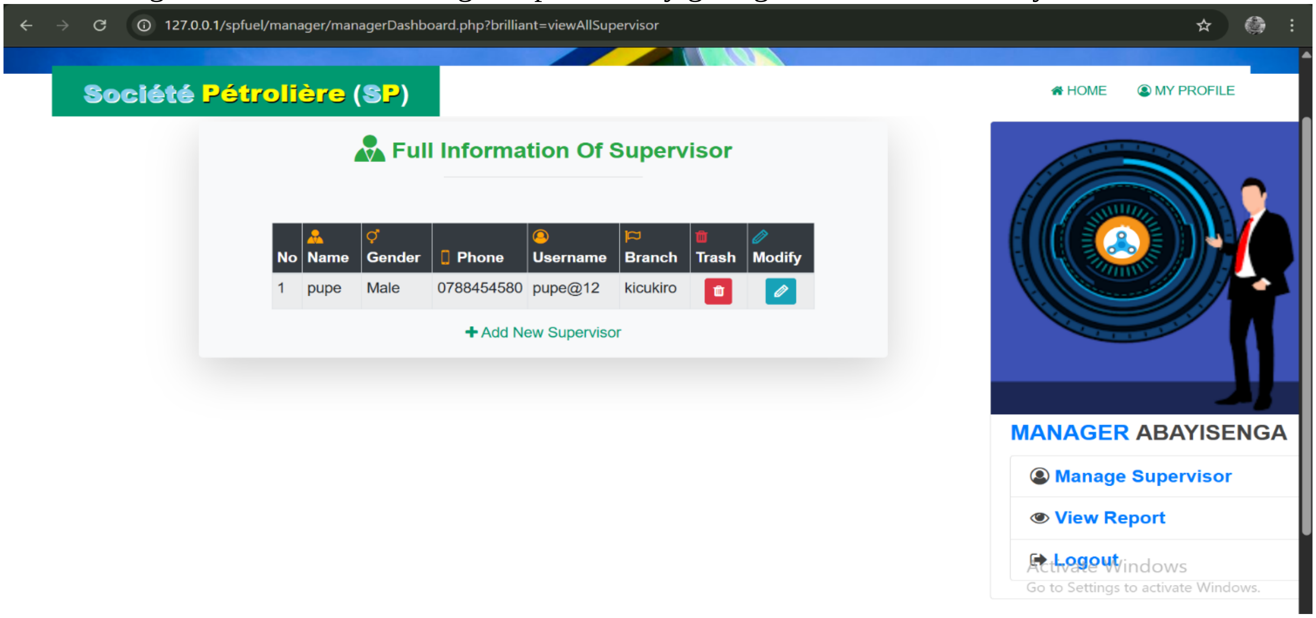


Figure 15: Supervisor allocation form

5.3.2 Manager also can view reports of fuels sold

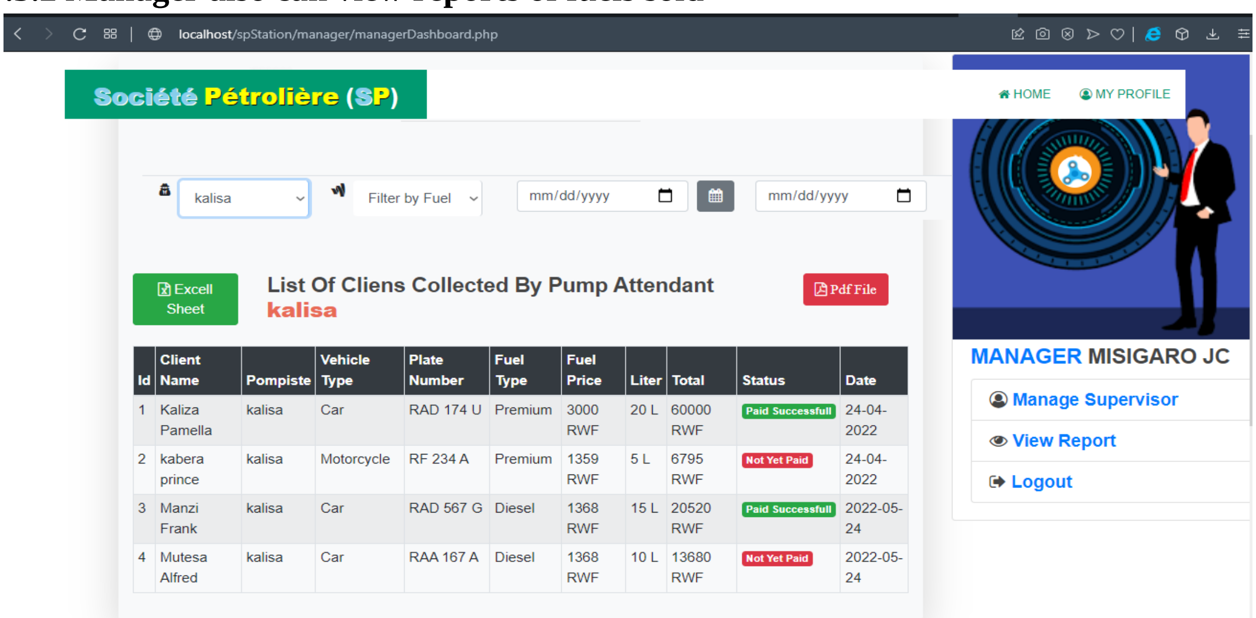


Figure 16: Manager Report form detail

5.4 Supervisor login form

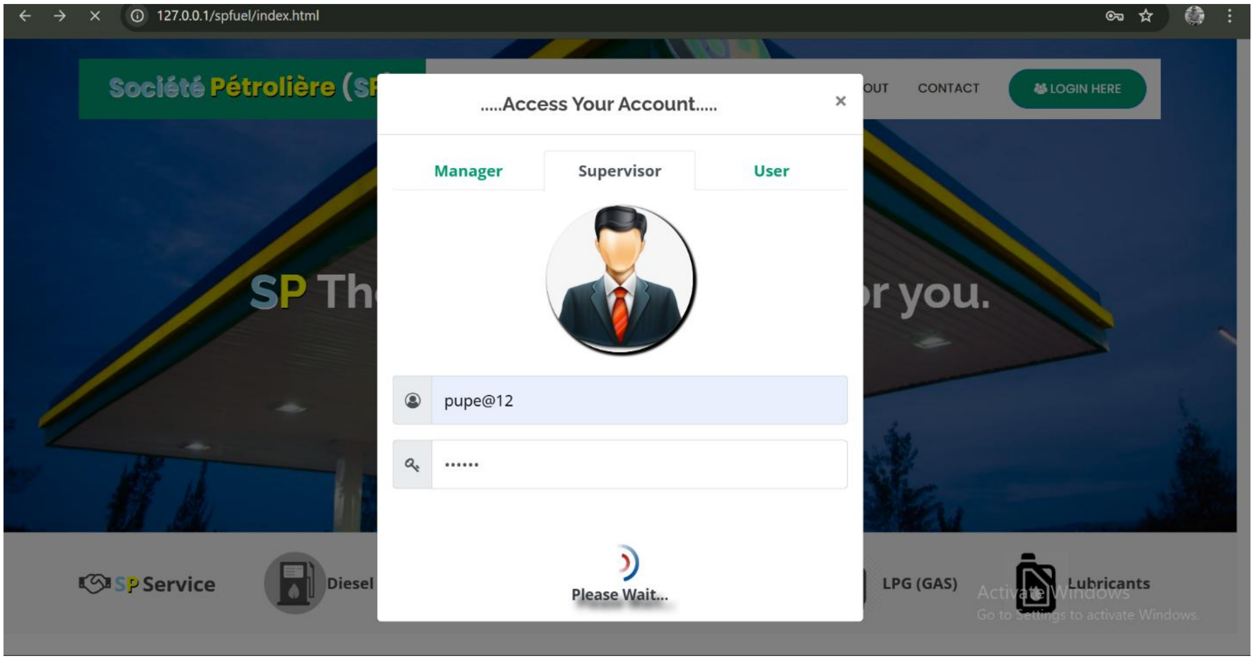


Figure 17: Supervisor login form

5.4.1 Supervisor dashboard

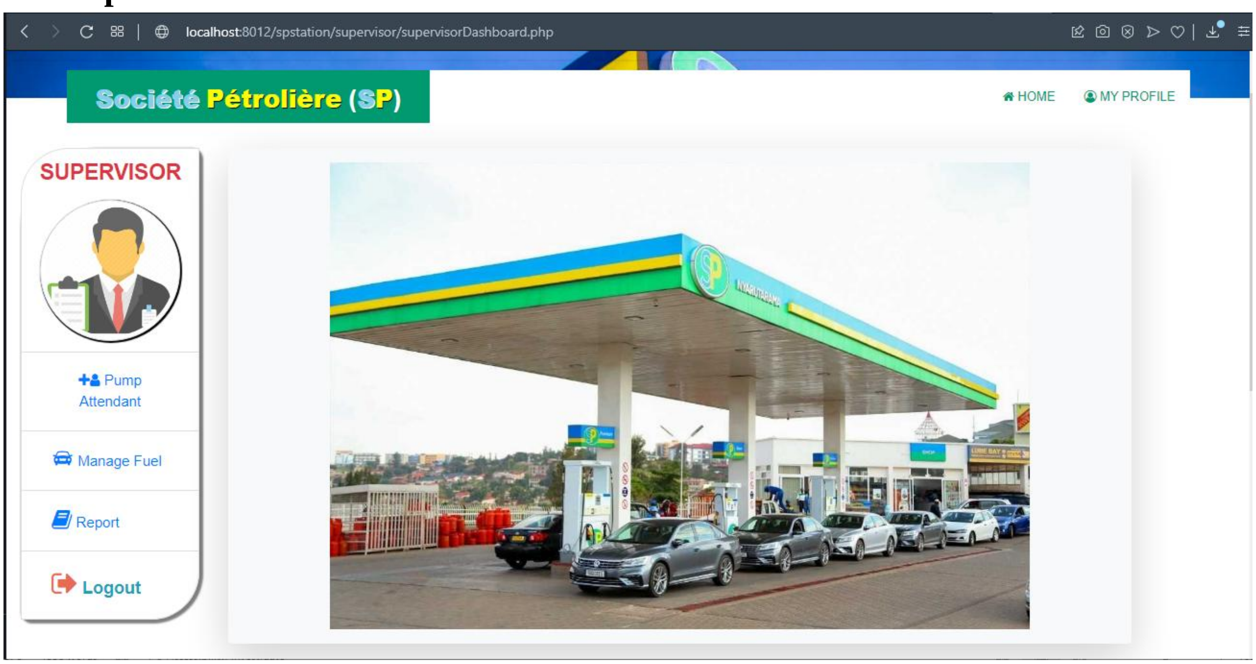


Figure 18: Supervisor dashboard

5.4.2 Supervisor adding pompiste credentials

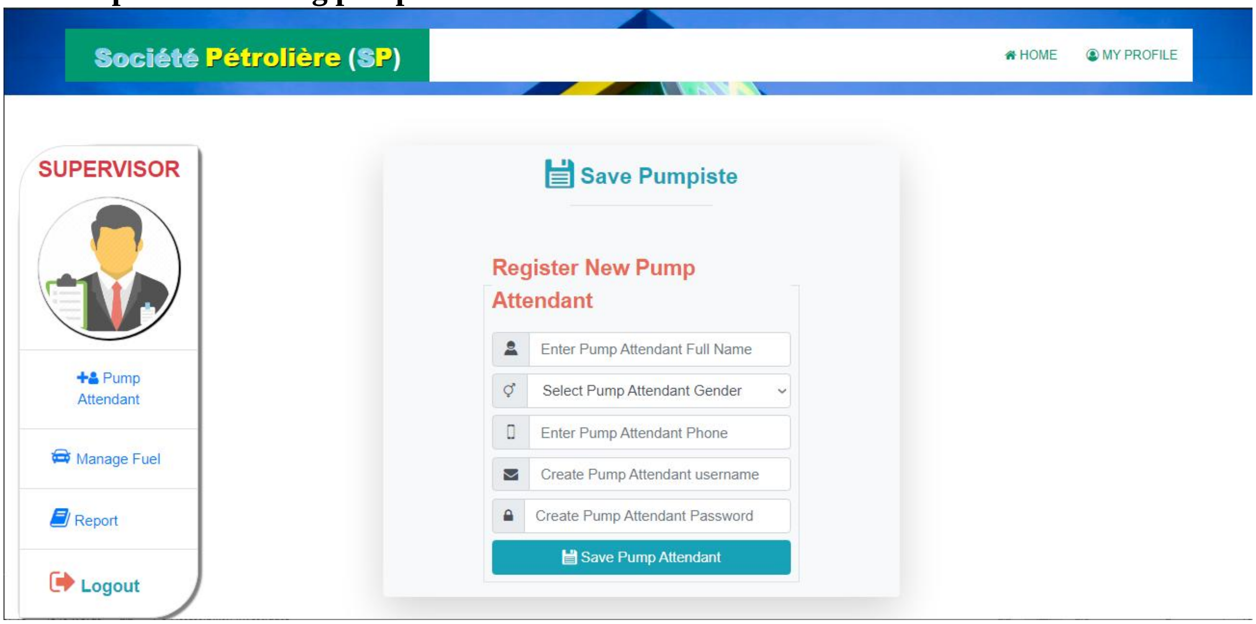


Figure 19: Pompiste allocation form

5.4.3 Fuel Management Form

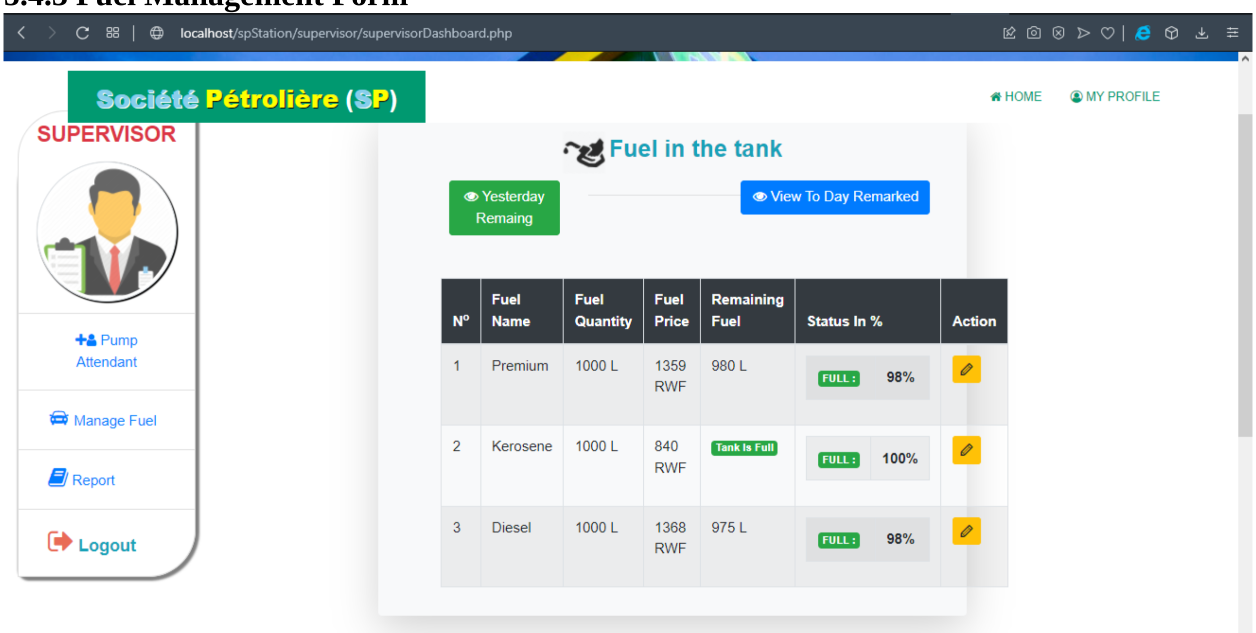


Figure 20: Fuel details form

5.4.4 Supervisor report management

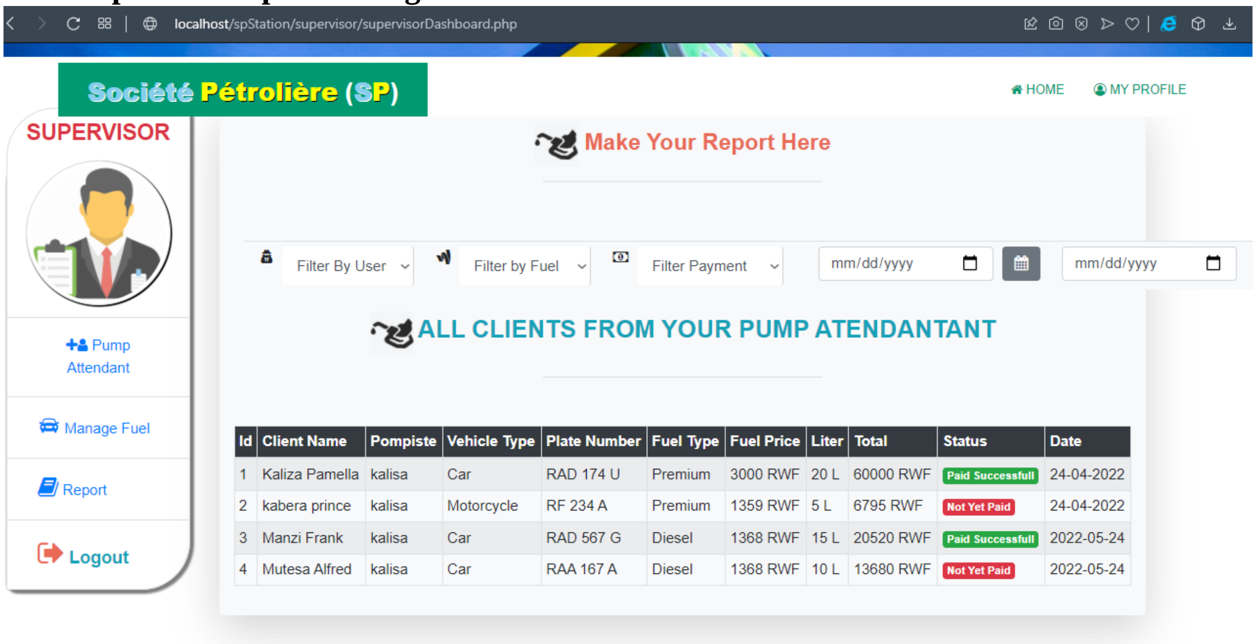


Figure 21: Report form details

5.5 Pompiste dashboard

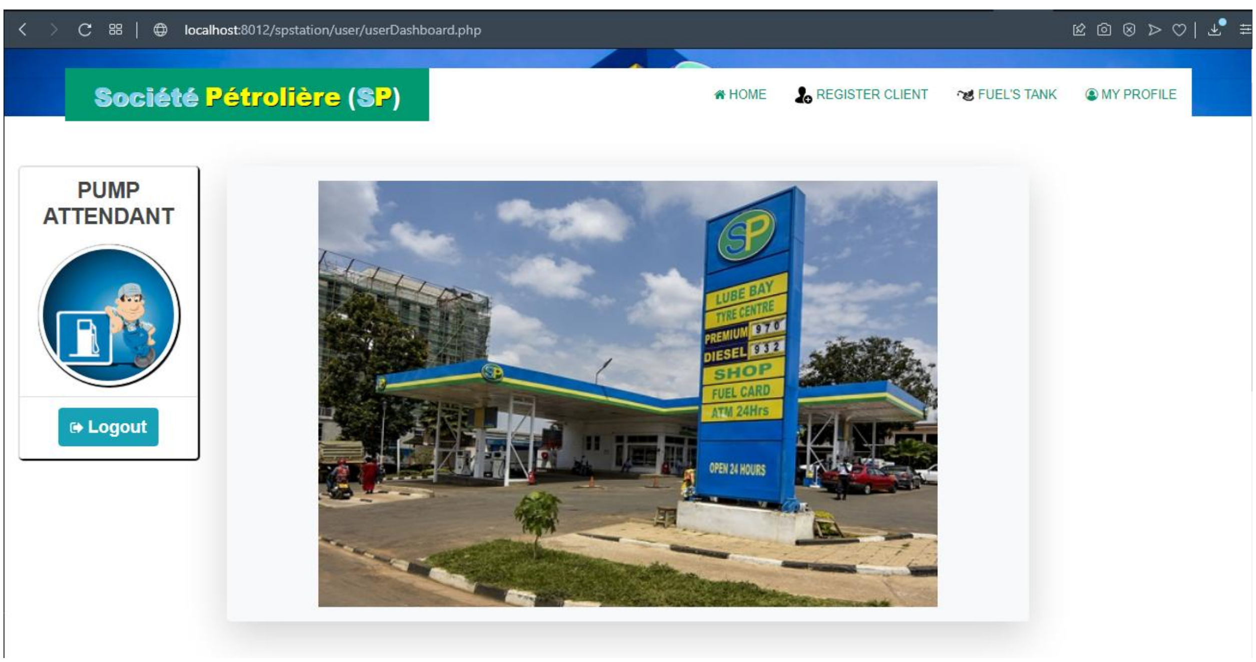


Figure 22: Pompiste Dashboard

5.5.1 Client information records by pompiste

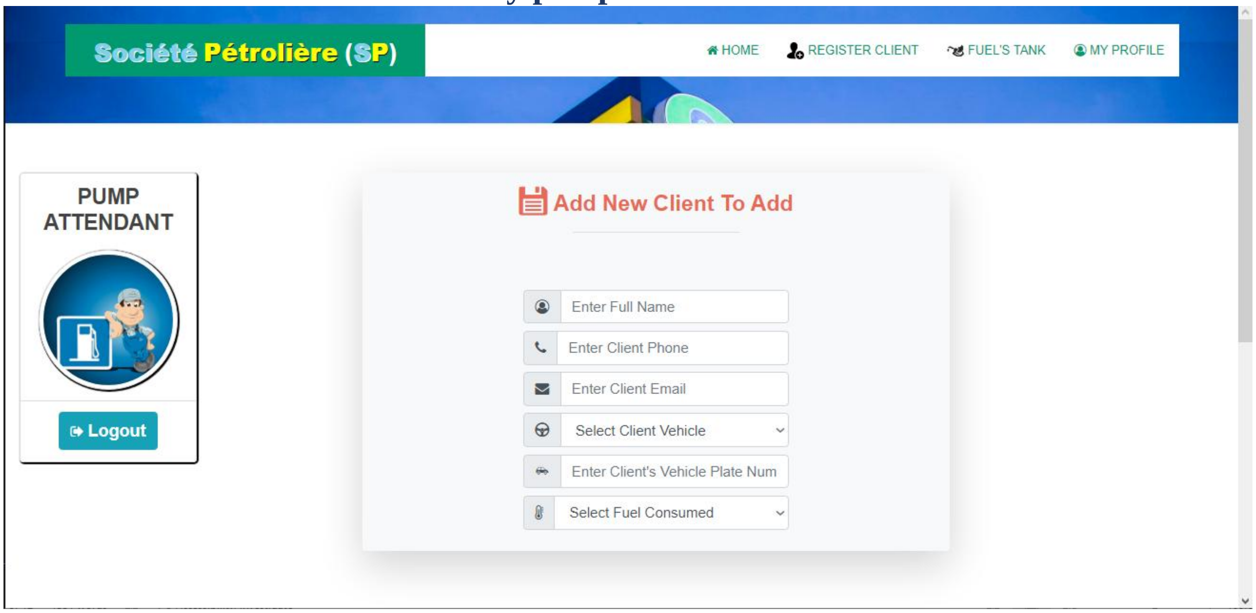


Figure 23: Pompiste Dashboard

5.5.2 Transactions record lists

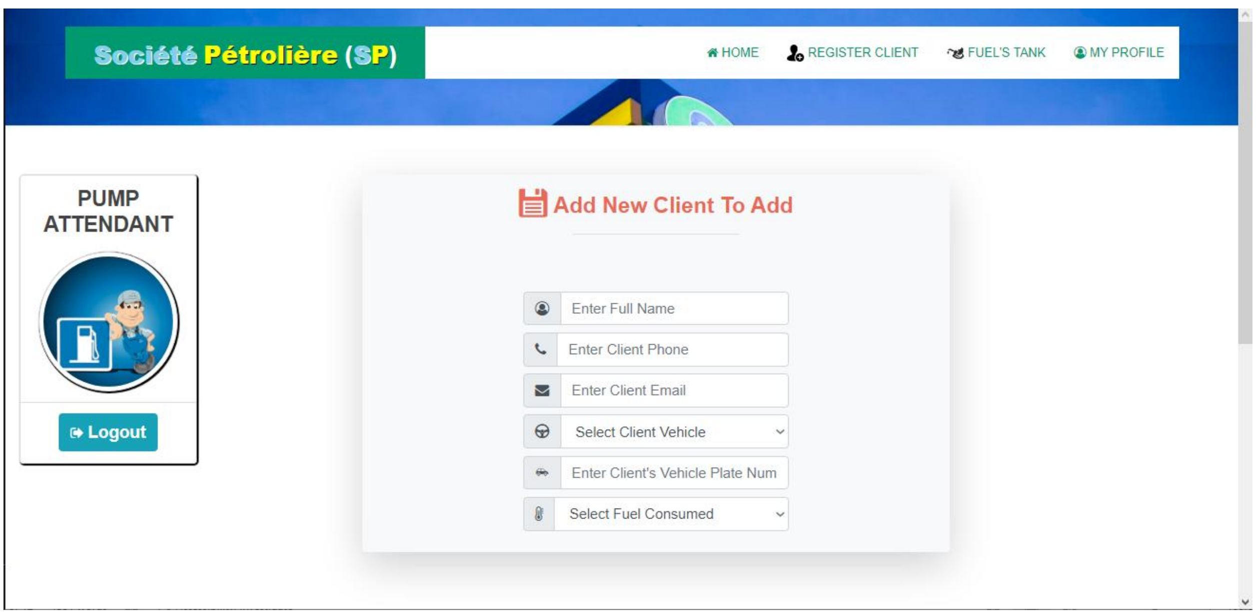


Figure 24:Transactions record lists

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5.5.3 Client details form

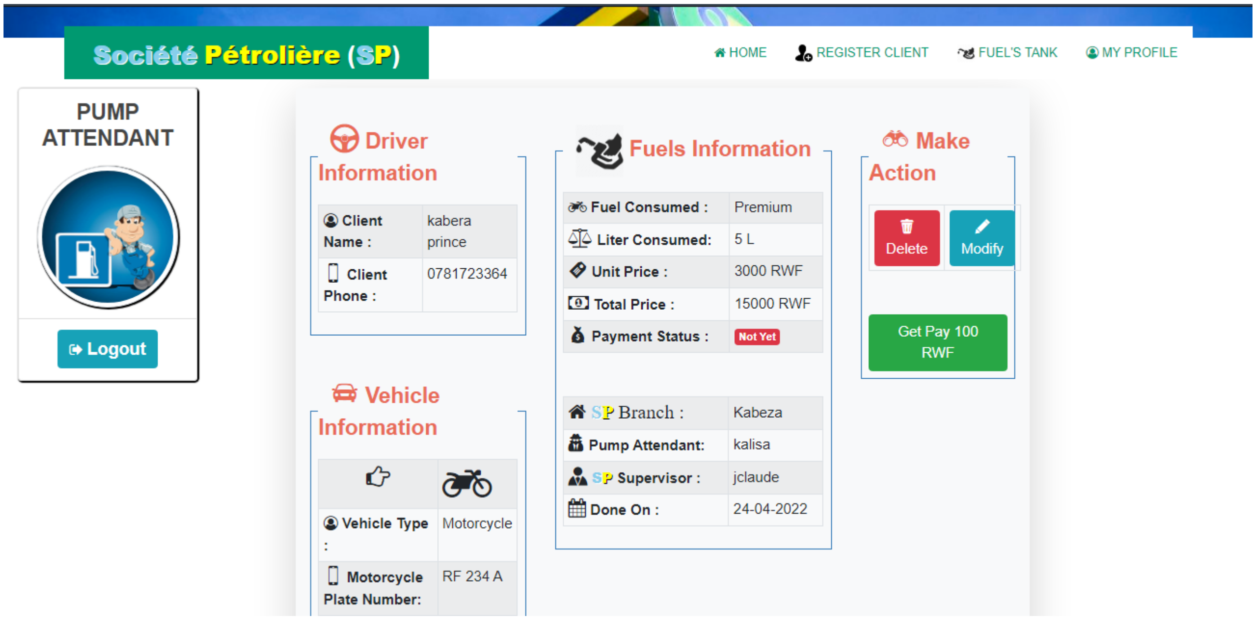

Figure 25: Clients details form

5.6 Software Technologies

Software tools are essential for developers to create, debug, and maintain applications. These tools include source code editors, compilers, and various programming languages suited for different tasks in software development.

5.6.1 Tools and Languages Used

Various tools and programming languages are utilized to build the Smart Petrol Station Management System, covering both server-side and client-side programming.

5.6.2 HTML

HTML (Hypertext Markup Language) is employed to create the web interface for the petrol station management system. It defines the structure of web pages, determining how text, images, and other elements are displayed in the browser.

5.6.2.1 HTML Tags Definition

HTML tags are instructions enclosed in angle brackets (< and >) that guide the browser in rendering content. Examples include <div>, <h1>, and <p>.

5.6.3 CSS

CSS (Cascading Style Sheets) is used to style HTML elements, allowing developers to separate content from design. This enhances the visual appeal and user experience of the petrol station management interface.

5.6.4 PHP

PHP (Hypertext Preprocessor) serves as the server-side scripting language for the system, enabling the creation of dynamic web pages. It processes transaction data, manages user sessions, and integrates with the RFID payment system for efficient transactions.

5.6.5 php MyAdmin

phpMyAdmin is a web-based tool written in PHP for managing MySQL databases. It simplifies tasks such as creating databases, managing tables, and executing SQL queries related to the petrol station's transaction records.

5.6.6 XAMPP Server

XAMPP is a free, open-source web server solution that includes Apache, MySQL, and interpreters for PHP and Perl scripts. It is easy to install and use, providing a local development environment for the Smart Petrol Station Management System.

5.6.7 Browser

A web browser is a software application used to access and display the web-based interface of the Smart Petrol Station Management System.

5.6.8 Static Website

A static website delivers the same content to all users, with no changes based on user interactions. This is not applicable to the petrol station system, which requires dynamic functionality.

5.6.9 Dynamic Website

A dynamic website adjusts its content based on user interactions and other factors, such as real-time fuel pricing and transaction status updates. The Smart Petrol Station Management System is a dynamic application designed for this purpose.

5.6.10 Hardware Technologies

This section outlines the hardware components necessary for the system's operation, including:

- RFID Card Readers: Essential for processing customer payments through RFID cards, enabling quick and secure transactions at the petrol station.
- Servers: To host the application and manage database transactions related to customer profiles and transaction records.
- Network Devices: Routers and switches to facilitate communication between components and ensure seamless data transfer during transactions.

5.7 System Testing

System testing is critical to ensuring the quality and functionality of the Smart Petrol Station Management System. It helps identify bugs and confirms that the software meets user expectations.

5.7.1 Objective of the Testing

The primary goal of software testing is to verify the quality of the product. Testing identifies bugs, ensuring that the software functions as intended and demonstrating its reliability and effectiveness.

5.7.2 Unit Testing

Unit testing involves checking individual components of the system to identify and fix errors at an early stage of development. This helps ensure that each part of the application, including the RFID transaction processing, works correctly before integration.

5.7.3 Validation Testing

Validation testing confirms that the system meets the defined requirements and performs its intended functions, particularly in processing RFID transactions and generating reports related to fuel sales and customer usage.

5.7.4 System Testing

System testing evaluates the entire Smart Petrol Station Management System to ensure that all components, including the RFID technology and database management, work seamlessly together to provide accurate transaction processing and reporting.

5.7.5 Testing Approaches

Different testing methodologies will be applied to ensure comprehensive coverage:

5.7.5.1 White Box Testing

White box testing examines the internal workings of the application, allowing testers to assess code efficiency and identify areas for improvement based on the logic and structure of the software, especially in transaction handling.

5.7.5.2 Black Box Testing

Black box testing focuses on the functionality of the Smart Petrol Station Management System without considering internal code structure. Testers will input values (e.g., RFID card data) and check if the outputs (e.g., transaction confirmations and receipts) match expected results.

CHAPTER 6: CONCLUSION AND RECOMMENDATIONS

6.1 Limitations

This chapter discusses the challenges encountered during the development of the Smart Petrol Station Management System, offers recommendations for improvements, and summarizes the key findings of the project.

6.2 Limitations

Several challenges were faced throughout the project, including:

1. **Information Gathering**: A significant issue was the difficulty in collecting comprehensive information, as some potential users were hesitant to share their experiences and requirements. This reluctance limited our understanding of user needs and expectations.

2. **Internet Dependency**: The system's reliance on internet access presents a limitation for users in areas with poor connectivity. This could hinder the functionality of the RFID transaction processing and overall system performance in these regions.

3. **Language Accessibility**: The current version is only available in English, which restricts accessibility for non-English-speaking users, particularly those who speak French or Kinyarwanda. This language barrier may exclude potential users from fully utilizing the system.

4. **Hardware Integration**: Challenges in integrating hardware components, such as RFID card readers and network devices, may affect system performance and user experience. Ensuring seamless communication between hardware and software is crucial.

6.3 Recommendations

While the Smart Petrol Station Management System is operational, several areas for improvement can enhance its effectiveness:

1. **Language Accessibility**: To improve usability and cater to a broader user base, it is recommended to develop multilingual support, including French and Kinyarwanda, ensuring that non-English speakers can effectively use the system.

2. **Database Scalability**: Updating the database structure to handle a larger number of users and transactions will ensure the system can grow without performance issues, particularly as more petrol stations adopt the technology.

3. **Security Enhancements**: As new features and functionalities are added, it is crucial to continuously enhance security measures to protect user data, especially regarding sensitive information processed during RFID transactions.

4. **Offline Capabilities**: Developing offline functionality can mitigate the dependency on internet access and enhance the system's reliability, allowing users to continue operations during connectivity issues.

5. **User Training and Support**: Offering training sessions and comprehensive user support will help users adapt to the new system and maximize its potential, particularly regarding the RFID payment process.

6.4 Conclusion

The primary objective of this project was to develop an efficient Smart Petrol Station Management System that utilizes RFID technology for transactions. This involved understanding current practices in fuel management and designing a system to address identified user needs. The outcome is a system that significantly improves transaction efficiency, data security, and user experience. Through various methods and tools, we analyzed and designed the system, resulting in a final product that meets specified requirements and provides a robust solution for managing petrol station operations effectively.

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APPENDICIES

FRONT END.

make-station-show.blade.php:

```
@extends('clerk.layouts.admin')
@section('css')

@section('content')

<div class="container pt-3 pt-md-4">
    <div class="row justify-content-center">
      <div class="col-12 col-lg-10 ">

        <!-- Header -->
        <div class="header mt-md-5">
          <div class="header-body">
            <div class="row align-items-center">
              <div class="col">

                <!-- Pretitle -->
                <h6 class="header-pretitle">
                  station
                </h6>

                <!-- Title -->




              </div>
              <div class="col-auto">
```

```
@if($x->owner==null)

<a href="{{route('clerk.edit.fuele,$x->id)}}" class="btn btn-light">edit</a>

@endif
{{-- @if($x->isAvailable=='1' and $x->owner!=null)
<a href="{{route('make-fuel',$x->id)}}" class="btn btn-light"

 onclick="event.preventDefault();
             document.getElementById('claim').submit();">
    claim
</a>

  <form id="claim" action="{{ route('clerk. edit.station ',$x->id) }}" method="POST"
style="display: none;">
  @csrf
</form>
 @endif --}}
 </div>
 @if($x->owner==null)

 @endif
</div> <! -- /. row -->
</div>
</div>
@include ('includes. Fuel)
</div>
</div>
</div>
```

@end section

BACK END.

ConfirmPasswordController:

```
<?php

namespace App\Http\Controllers\Auth;

use App\Http\Controllers\Controller;
use App\Providers\RouteServiceProvider;
use Illuminate\Foundation\Auth\ConfirmsPasswords;

class ConfirmPasswordController extends Controller
{
    /*
    |--------------------------------------------------------------------------
    | Confirm Password Controller
    |--------------------------------------------------------------------------
    |
    | This controller is responsible for handling password confirmations and
    | uses a simple trait to include the behavior. You're free to explore
    | this trait and override any functions that require customization.
    |
    */

    use Confirms Passwords;
```

```
    /**
     * Where to redirect users when the intended url fails.
     *
     * @var string
     */
    protected $redirectTo = RouteServiceProvider: HOME;

    /**
     * Create a new controller instance.
     *
     * @return void
     */
    public function __construct ()
    {
        $this->middleware('auth');
    }
}
```